Mathematical overview: facts, assumptions and limits

At the printed crossing, read from top to bottom. The left incoming color a passes under, and the right incoming color b passes over to the left output. The right output has color c uniquely determined by the all-same/all-different rule. With R/B/G encoded 0/1/2, (a,b) maps to (b,2b-a) modulo 3. Equal input pairs are fixed; unequal pairs have exact period 3. This is a statement about left/right **positions**, not named cord identities: each crossing exchanges the two positions. A whole arc continues through an overpass, and an underpass separates its incoming and outgoing arcs.

Closing the two-strand strip joins left output to left input and right output to right input, without new crossings. Valid colorings correspond exactly to fixed ordered pairs of the n-crossing machine. Counts for n=1,2,3,4,6 are respectively 3,3,9,3,9. There is one separate cord for odd n, two for even n. Thus this coloring count can distinguish some diagrams but misses cord/component distinctions for others. Counts include all named-color choices and the three constant colorings; rotating the paper or renaming colors does not identify assignments.

For the printed three-crossing loop, the open three-crossing machine is identity. Cutting its crossing-free outside return arc leaves ends that necessarily have the same color in every valid coloring. For each fixed cut color there are 3 valid colorings of the rest. The plain loop has 1 per fixed cut color. Joining two cut models with matching end colors therefore gives $3\times3\times3=27$ for two three-crossing loops, and $3\times3\times1=9$ for one plus a plain loop. Three three-crossing loops give 81. More generally the product/divide-by-three count applies when each chosen cut diagram is end-monochromatic and named colors are distributed equally at the distinguished cut; those assumptions hold for the delivered examples, as checked by the machine. The guide does not assert that an arbitrary open tangle has matching end colors. Joining a cut arc does not mean blindly reducing the number of diagram arcs by one: count whole gap-to-gap arcs in the actual joined diagram.

Children discover periodic color propagation, closure restrictions, and composition of counts. Grades 2–3 can enter the open machine after reliable whole-arc tracing; Grades 4–5 can explain complete counts and predict compositions. No modular arithmetic, knot classification or undergraduate prerequisites are required. A coloring experiment is evidence; propagation/closure arguments establish exact counts. Equal counts do not establish knot equivalence, and these pages make no unsupported claim about which closed pictures can be untangled.

Materials, readiness, launch and visits

Per pair: one copy of the three student pages at 100%, two transparent page protectors or reusable tracing sheets, three colored pencils R/B/G (fine enough for the 5 pt=1.76 mm white channel), pencil/eraser, nine input-pair cards RR/RB/RG/BR/BB/BG/GR/GB/GG, one separate L/R current-position mat with two 25 mm circles, and six counters of each color. Use letters/patterns alongside colors. The partner keeps the original input card fixed, checks each crossing, and moves the current pair between the L/R sites. This externalizes the changing state instead of making the whole rule mental. Counters do not go in the narrow diagram channel. For six full pair kits across K–1 / 2–3 / 4–5: six packets, 12 protectors, 18 colored pencils, 54 input cards, six L/R mats, 36 counters of each color. Practical active stock can be three older-table kits; no K–1 version is forced here. K–1 may revisit the base cord/arc gate if suitable. Three anchored adults remain sufficient for the selected readiness routes.

Prepare all cards/mats before class; no child cord cutting is required. These are paper color investigations. Physical cut-and-join cord models are optional adult-prepared demonstrations and are **not supplied or certified** by the rendered diagrams. Reuse any base cord only after its own before-use checks. For paper coloring, exact marker/pencil fit and children's arc tracing have **not** been physically pretested. Do not substitute thick markers without checking their fit.

Launch together: handle the input counters and trace a true over/under crossing. Use the printed non-task R/B example: R comes from top left, B from top right passes over, so output left is B and output right must be G. Point to the meaningful intermediate over-B strand, then to the ordered output. Do not switch strands at the projected intersection. Keep the input card while the partner operates/checks the machine. Before P2, trace each outside closure from an output back to the

same-position input. Before P3, use the plain-loop input/cut/join/output visual; its facing gray pieces are the marked cuts: cuts lie away from crossings; upper ends join together and lower ends together. The two joining segments are part of the final whole loop, not an extra road or crossing.

A first visit can work P1 for 20–30 minutes, then one/two P2 diagrams for 15–25. Offer the remaining closures and P3 as return visits of 20–40 minutes each; complete enumeration/proof may use several visits. Stop after a complete count with a defensible reason. Gates: each child must trace through an overpass, distinguish a gap, and apply the crossing rule to a near miss before operating the machine. Adult reading is routine; adults should not choose all missing colors for children. Timing, physical pencil fit, counters and child transfer remain unpiloted and untested.

Solutions, hints and deeper work

P1 / student p.1. RR,BB,GG first return after 1 crossing. Each of RB,RG,BR,BG,GR,GB first returns after 3. No pair fails to return. The six unequal pairs split into two cycles:

RB -> BG -> GR -> RB

RG -> GB -> BR -> RG

Input a=b gives c=b=a. If a≠b, the output uses the remaining third color in the right position; iterating gives each ordered pair above. Save one current pair and one fixed input card; only the first-return number needs recording. To explain complete coverage, check the three equal inputs and both three-cycles. Hint: physically track left/right positions, rather than try to follow a single cord identity around the machine. Extension: build a machine with crossings read backward and compare its inverse action, or ask what repeating a three-crossing block can change at the boundary. These are continuations of the machine investigation, not additional counted activities.

P2 / student p.2. In printed order:

Crossings	Total colorings	Separate cords
1	3	1
2	3	2
3	9	1
4	3	2
6	9	2

A choice of the two top-position colors forces every remaining arc by successive crossings. It closes exactly when the bottom pair matches the top pair componentwise. If n is divisible by 3, all 9 pairs work. Otherwise only equal pairs work, giving 3. To count cords, each crossing exchanges positions; an odd number makes the two outside returns part of one trip, an even number gives two separate trips. One, two and four crossings share count 3 but not cord count; three and six share count 9 but not cord count. Do not count total projected loops or enclosed regions. Hint: keep one named input pair and test its actual return, then decide how many input cards meet the condition. Extension: compare n=5 and n=9 using the rule; discuss whether equal counts can decide knot equivalence. The base theorem says differing counts obstruct equivalence; equality is inconclusive.

P3 / student p.3. The top joined picture has 27 total colorings. The lower has 9. Three three-crossing loops joined have 81. For one trefoil cut at its right outside return, fix its cut color b: the other top color a has 3 choices, each uniquely extends. The right trefoil cut at its left outside return likewise has 3 choices when its cut color matches b. Thus for each of 3 joined colors there are 3×3=9 combinations, 27 overall. A cut plain loop has no further choice, so the lower count is 3×3×1=9. A third trefoil multiplies the total by 3, giving 81. A child may pool cases by the joined color, saving two local color choices rather than redrawing 27 entire diagrams. Verify upper and lower joined segments agree as forced by the three-crossing identity; do not assume they agree just because the same color pencil was used in one trial.

The drawing has six actual crossings in the top target, three in the lower, and one connected cord in each. Each trefoil is the same-handed closure of three identical crossings; the joining strands introduce no new crossings. This is a composition calculation, not a proof of connected-sum uniqueness, knot primeness or the full Reidemeister theorem. Hint: fix the color on a crossing-free joining segment and ask how many local choices remain on each side. Extension: choose another distinguished arc in one model and verify the count stays the same for that specified construction; include a plain loop or add another three-crossing loop. Retain explanations alongside experiments.

Provenance, checks and limits

Base Week 40 compares a trefoil/plain loop, checks a type-II move, and states the coloring invariant's one-way obstruction. New P1 is an open boundary machine; P2 closes a family and contrasts color count with cord count; P3 composes cut diagrams. The diagrams are original elementary braid renderings, not copied knot images. Existing Week 40 facilitator source notes supply the all-same/all-different context. Math Circle by the Bay, preface printed pp. viii–x (PDF pp. 9–11), was read: it recommends deep themes, manipulatives, spare challenging problems and repeated improvement of explanations, and expects five large themes to take at least two years. Our separate encore visits are a planning inference; the source says they almost never repeat the same topic for the same children.

student/verify.py checks all 9 crossing completions and all input return periods; independently enumerates local all-same/all-different choices; checks all printed closure colors; computes cord counts both from position cycles and actual centerline endpoint connectivity; counts both cut/join targets and the 81-color continuation; and checks exact global centerline intersections so there are no unrecorded extra crossings. The final renderer merges endpoint-connected pieces into each continuous gap-to-gap arc before drawing its black outline and white channel; only genuine underpass gaps and open ends interrupt the channel. Build assertions check eight arcs in the six-crossing open machine, n arcs in each n-crossing closure, and six/three arcs in the joined targets. The finite algebra, graph and rendered continuity checks do not certify physical pencil fit, models or classroom readiness. Portable student/build.sh OUTPUT_DIR requires Python 3 and ReportLab and uses no repository imports. Record actual use as W40-B1/B2/B3. All outputs are unpiloted; physical pretests remain unperformed.